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HAND FOLD TRAYS

Abstract

A carton for storage and display of articles can be manually formed from a blank and includes a bottom wall of the blank, a first side wall foldably connected to the bottom wall, a second side wall disposed opposite the first side wall and foldably connected to the bottom wall, a first end wall foldably connected to both the bottom wall and to the first and second side walls, and a second end wall disposed opposite the first end wall and foldably connected to both the bottom wall and to the first and second side walls. One side of the blank defines a continuous surface partially defined by interior-facing surfaces of the first and second side walls, and exterior-facing surfaces of the first and second side walls.

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Background/Summary

BACKGROUND

1. Field

[0001] The present invention relates to a carton, to a blank, a method for forming the carton, and more particularly to a carton formed from a blank having a first side and a second side, the first side defining the interior surface of the carton's bottom wall and the interior facing surfaces and the exterior facing surfaces of the carton's side walls where the first side includes a marking and the second side does not.

2. Description of Related Art

[0002] Reusable plastic crates (RPCs) are widely-used in the grocery industry for transporting, storing, and distributing foods such as fresh fruits and vegetables. Advantages of RPCs include a stackable interlocking design, a “display ready” design that can significantly reduce stocking time, and a foldable design that reduces storage space and transport costs by allowing the containers to be folded when not in use. Disadvantages of RPCs include the risk of loss/theft and a relatively high cost resulting from the need to wash and sanitize the RPCs after each use and transport them to a specialized facility for that purpose.

[0003] Accordingly, there is a need in the art for a carton configuration having the stackability and branding opportunities of an RPC but at the same time being economically viable. The present disclosure may provide a solution for at least one of these remaining challenges.

SUMMARY

[0004] A carton for storage and display of articles includes a bottom wall of a blank, a first side wall foldably connected to the bottom wall, a second side wall disposed opposite the first side wall and foldably connected to the bottom wall, a first end wall foldably connected to both the bottom wall and to the first and second side walls, and a second end wall disposed opposite the first end wall and foldably connected to both the bottom wall and to the first and second side walls. Each of the bottom wall, the first and second side walls, and the first and second end walls are configured to partially define an interior space. One side of the blank defines a continuous surface partially defined by an interior-facing surface of the first side wall, an interior-facing surface of the second side wall, and an exterior-facing surface of the first side wall, an exterior-facing surface of the second side wall.

[0005] The continuous surface can be partially defined by an interior-facing surface of the bottom wall. The first end wall can include a first end panel connected along a fold line to the bottom wall, and a first end flap connected along a fold line to the first end panel. An exterior-facing portion of the first end panel can be defined by a surface opposite the continuous surface. An interior-facing portion of the first end flap can be defined by the surface opposite the continuous surface. End wall flaps of the first and second side wall can be layered between the first end panel and the first end flap. One or more locking tabs can extend from an end of the first end flap, wherein the one or more locking tabs is engaged in one or more respective locking slots in the bottom wall.

[0006] The continuous surface can be partially defined by an exterior-facing surface of the bottom wall. The first end wall can include a first end panel connected along a fold line to the bottom wall, and a first end flap connected along a fold line to the first end panel. An exterior-facing portion of the first end panel can be defined by the continuous surface, and wherein an interior-facing portion of the first end flap is defined by the continuous surface.

[0007] End wall flaps of the first and second side wall can be layered between the first end panel and the first end flap. One or more locking tabs can extend from an end of the first end flap, wherein the one or more locking tabs is engaged in one or more respective locking slots in the bottom wall.

[0008] A first corner wall can be foldably connected to the first side wall, and to the first end wall. A second corner wall can be foldably connected to the first end wall, and to the second side wall. A third corner wall can be foldably connected to the second side wall, and to the second end wall. A fourth corner wall can be foldably connected to the second end wall, and to the first side wall. The continuous surface can further include an interior-facing surface of the first corner wall, an interior-facing surface of the second corner wall, an interior-facing surface of the third corner wall, and an interior-facing surface of the fourth corner wall. The continuous surface can further include an exterior-facing surface of the first corner wall, an exterior-facing surface of the second corner wall, an exterior-facing surface of the third corner wall, and an exterior-facing surface of the fourth corner wall.

[0009] The continuous surface can be a treated surface and a second surface, opposite the continuous surface is an untreated surface. The treated surface can include one or more surface treatments. The untreated surface need not include the one or more surface treatments.

[0010] A blank for forming a carton includes a first surface, a second surface disposed opposite the first surface, a plurality of panel and flaps configured to form a carton defining a bottom wall, a first side wall foldably connected to the bottom wall, a second side wall disposed opposite the first side wall and foldably connected to the bottom wall, a first end wall foldably connected to the bottom wall, a second end wall disposed opposite the first end wall and foldably connected to the bottom wall. Each of the bottom wall, the first and second side walls, and the first and second end walls are configured to partially define an interior space. The first surface is configured to be at least partially defined by an interior-facing surface of the bottom wall, an interior-facing surface of the first side wall, an interior-facing surface of the second side wall, an interior-facing surface of the first end wall, an interior-facing surface of the second end wall, an exterior-facing surface of the first side wall, an exterior-facing surface of the second side wall, an exterior-facing surface of the first end wall, and an exterior-facing surface of the second end wall in a carton formed from the blank.

[0011] The plurality of flaps can include a first side wall panel foldably connected to a bottom panel, and a first side flap foldably connected to the first side wall panel, wherein each of the first side wall panel and the second side wall panel include respective laterally extending corner panels foldably connected thereto. Each of the respective laterally extending corner panels can include a respective end panel foldably connected thereto.

[0012] A method of forming a carton includes starting with a flat blank with a first treated surface on one side and a second surface on an opposite side from the first treated surface, manually forming a container from the flat blank. Manually forming the container includes folding a first side wall panel upward along a fold line connecting the first side wall panel to a bottom panel, folding a second side wall panel downward along a fold line connecting the second side wall panel to the first side wall panel to form a first side wall with an interior-facing surface of the first side wall and an exterior facing surface of the first side wall both formed of the first treated surface, folding a lateral end flap of at least one of the first side wall panel and/or second side wall panel to form a portion of a first end wall, and folding a first end panel along a fold line connecting the first end panel to the bottom panel and folding a first end flap along a fold line connecting the first end flap to the first end panel to enclose the lateral end flap between the first end panel and the first end flap. The method includes securing the first end flap to the bottom panel to form the first end wall.

[0013] Manually forming the container from the blank can include forming the bottom panel into a bottom wall of the container wherein the first treated surface is on an inward facing surface of the bottom wall. Inward and outward facing surfaces of the first end wall can be formed of the second surface of the blank.

[0014] Manually forming the container from the blank can include forming the bottom panel into a bottom wall of the container wherein the second surface is on an inward facing surface of the bottom wall. Inward and outward facing surfaces of the first end wall can be formed of the first

treated surface.

[0015] These and other features of the systems and methods of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description of the preferred embodiments taken in conjunction with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, preferred embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

[0017] FIG. 1 is a schematic perspective view of an embodiment of a container constructed in accordance with the present disclosure, showing the interior and exterior facing surfaces of the first and second sidewalls formed from a first surface of the blank;

[0018] FIG. 2 is a plan view of the blank for forming the container of FIG. 1;

[0019] FIGS. 3-5 are schematic perspective views of the blank of FIG. 2, schematically indicating folding of the blank to form the container of FIG. 1;

[0020] FIG. 6 is a schematic perspective view of an embodiment of a container constructed in accordance with the present disclosure, showing the interior and exterior facing surfaces of the first and second sidewalls formed from a first surface of the blank and the interior and exterior surfaces of the end walls are formed of a second surface of the blank;

[0021] FIG. 7 is a plan view of the blank for forming the container of FIG. 6; and

[0022] FIGS. 8-10 are schematic perspective views of the blank of FIG. 7, schematically indicating folding of the blank to form the container of FIG. 6.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0023] Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, a partial view of an embodiment of a container or carton in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character **100**. Other embodiments of systems in accordance with the disclosure, or aspects thereof, are provided in FIGS. 2-10, as will be described. The systems and methods described herein can be used to manually form containers from blanks, wherein a single surface of the blank forms interior and exterior facing surfaces of certain walls of the container.

[0024] The carton **100** for storage and display of articles such as produce or the like, includes a bottom wall **102** corresponding to a bottom panel **104** of a blank **101** (shown in FIG. 2). A first side wall **106** is foldably connected to the bottom wall **102** along a fold line **110** (labeled in FIG. 2). A second side wall **108** is disposed opposite the first side wall **106** and is foldably connected to the bottom wall **102** along a fold line **112**. A first end wall **114** is foldably connected to both the bottom wall **102** along fold line **118** and to the first and second side walls **106**, **108** along fold lines **120**. A second end wall **122** is disposed opposite the first end wall **116** and is foldably connected to both the bottom wall **102** along a fold line **124** and to the first and second side walls **106**, **108** along fold lines **126**.

[0025] Each of the bottom wall **102**, the first and second side walls **106**, **108**, and the first and second end walls **116**, **122** are configured to partially define an interior space **128**. One side of the blank **101** (shown in FIG. 2) for forming the container **100** in FIG. 1 defines a continuous surface **130** partially defined by an interior-facing surface of the first side wall **106**, an interior-facing surface of the second side wall **108**, an exterior-facing surface of the first side wall **106**, an exterior-facing surface of the second side wall **108**. The exterior-facing surfaces of the first and second end

walls **116**, **122** are shaded in FIG. 1 because they are formed of the opposite surface **132** of the blank **101** from the continuous surface **130**, i.e. the opposite surface **132** is the face not shown in FIG. 2, but indicated with a broken lead line. All parts of the continuous surface **130** are connected to the whole of that surface by fold lines, score lines, perforations, or the like. The continuous surface **130** is not shaded in FIG. 1, and the opposite surface **132** is shaded. The continuous surface **130** can be a treated surface and the opposite surface **132**, can be an untreated surface, i.e. the surface **130** can include at least some treatment that the opposite surface **132** does not include the one or more surface treatments, e.g. wherein the one or more surface treatments comprises printing, coloring, or the like.

[0026] The first side wall **106** defines at least one first stacking tab **134** that projects from an upper edge of the first side wall **106** (as oriented in FIG. 1). The second side wall **108** similarly defines at least one second stacking tab **134** that projects from an upper edge of the second side wall **108** as oriented in FIG. 1. Each of the folding tabs **134** is formed of two folding tab portions of the blank **101** of FIG. 2, wherein the two portions are folded back onto the opposite surface **132**. The carton **100** defines at least one first side aperture **136** below the stacking tab(s) **134**, the defined in part by both the bottom wall **102** and the respective side wall **106**, **108**. Each side aperture **136** can be defined in part by both the bottom wall **102** and a respective one of the side walls **106**, **108**. Each side aperture **136** is configured to receive a respective stacking tab **134** of a second carton **100** for stacking of multiple cartons **100**.

[0027] The first end wall **116** includes a first end panel **138** connected along a fold line **118** to the bottom wall/panel **102**, **104**. The first end panels **138** of both end walls **116**, **122** are labeled in FIG. 2. A first end flap **140** is connected along a fold line **142** to the first end panel **138** in each of the end walls **116**, **122**, wherein both end flaps **140** of the end walls **116**, **122** are labeled in FIG. 2. In the carton **100**, an exterior-facing portion of the first end panel **138** in each end wall **116**, **122** is defined by a the opposite surface **132**. An interior-facing portion of each end flap **140** is also defined by the opposite surface **132**. End wall flaps **142** at each end of the first and second side walls **106**, **108** are layered between the end panel **138** and the end flap **140** of each end wall **116**, **122**. One or more locking tabs **144** (labeled in FIG. 2) extend from an end of each end flap **144**, wherein each locking tab is engaged in a respective locking slot **146** (labeled in FIG. 2) in the bottom wall/panel **102**, **104** for manually forming the carton **100** from the blank **101**, e.g. locking the carton as shown in FIG. 1 without the need for adhesive.

[0028] A first corner wall **146** is foldably connected to the first side wall **106** and to the first end wall **116** along a pair of the fold lines **120**. A second corner wall **148** is foldably connected to the first end wall **116**, and to the second side wall **108** along a pair of the fold lines **120**. A third corner wall **150** is foldably connected to the second side wall **108**, and to the second end wall **122** along a pair of the fold lines **120**. A fourth corner wall **152** is foldably connected to the second end wall **122**, and to the first side wall **106** along a pair of the fold lines **120**. This gives a perimeter of the container **100** defined by the first sidewall **106**, second sidewall **108**, first end wall **116**, second end wall **112**, and the corner walls **146**, **148**, **150**, **152** an octagonal shape. The continuous surface **130** includes the interior-facing and exterior facing surfaces the corner wall **146**, **148**, **150**, **152**. Each of the corner walls **146** is formed of a pair of corner panels **154** labeled in FIG. 2, between two respective fold lines **120**.

[0029] Each of the first and second side walls **106**, **108** is formed of back to back first and second side wall panels **156**, **158** (labeled in FIG. 2) foldably connected to one another along a respective fold line **160**, wherein each first side wall panel **156** is foldably connected to the bottom wall/panel **102/104** along a fold line **112**. A first side flap **142** is foldably connected to the each of the first and second wall panels **156**, **158**, along a fold line **120** wherein each of the side flaps **142** is connected to a respective one of the first and second side wall panels **156**, **158** by way of a respective corner panel **154** and fold lines **120**. The corner panels **154** and side flaps **142** extend laterally from their respective side wall panels **156**, **158** with a slightly oblique convergence angle θ , labeled in FIG. 2

for one of the corresponding pairs of the side flaps **142**, relative to the respective fold lines **160**, **112** to facilitate forming the container **100** and to assist with stacking tab alignment as described above with reference to stacking tabs **134**.

[0030] Referring now to FIG. 3, a method of forming the carton **100** includes starting with a flat blank as shown in FIG. 2, with the first treated, continuous surface **132** on one side, e.g. facing upward as oriented in FIG. 3. This process can be performed manually, forming a container **100** from the flat blank **101** of FIG. 2. As shown in FIG. 3, the a first side wall panels **156** are folded upward as oriented in FIG. 3 along the fold lines **112**. And the second side wall panels **158** are folded downward along the fold line **160** to form the side walls **106**, **108** shown in FIG. 1. The folding of the first and second side wall panels **158** is indicated in FIG. 3 by the large arrows labeled with reference character **1**. As shown in FIG. 4, with the side walls **106**, **108** formed, the corner panels **154** and lateral end flaps **142** can be folded along the fold lines **120** (labeled in FIG. 2), as indicated by the curved arrows labeled with reference character **2** to form inner portions of the end walls **116**, **122**. As shown in FIG. 5, with the inner portions of the end walls **116**, **122** formed, each of the end panels **138** can be folded upward as oriented in FIG. 5 along the fold lines **118** and the end flaps **140** can be folded downward over the end flaps **142** along a doubled fold line **162** connecting the first end flaps **142** to their respective first end panels **138** to enclose the lateral end flaps **142** between the respective first end panel **138** and the first end flap **148**. This includes securing the end flaps **140** to the bottom panel/wall **104/102**, e.g. using the tabs **144** in the slots **146**, to form the end walls **116**, **122** and lock the container **100** in the erected state of FIG. 1. This results in the bottom panel **104** of the blank **101** forming the bottom wall **102** of the container **100** wherein the treated surface **130** is on an inward facing surface of the bottom wall **102**. This also results in the inward and outward facing surfaces of the end walls **116**, **122** being formed of the opposite surface **132** of the blank **101**.

[0031] With reference to FIGS. 6 and 7, another configuration of the blank **101** and container **100** are shown, similar on other aspects to the blank **101** and container **100** of FIGS. 1-2. However, in FIGS. 6 and 7, the continuous surface **130** is partially defined by an exterior-facing surface of the bottom wall **102**, i.e. the interior facing surface of the bottom wall **102** is formed of the opposite surface **132**, i.e. the non-treated surface. Also, in FIGS. 6 and 7, the first end walls **116**, **122** in this configuration include exterior-facing and interior facing surfaces defined by the continuous surface **130**. Manually forming the container **100** from the blank **101** in FIGS. 6-7 is shown in FIGS. 8-10. This includes starting with the opposite (e.g. untreated) surface **132** facing upward as in FIG. 8, so the bottom panel **104** of the blank **101** ultimately forms into the bottom wall **102** of the container **101** with the surface **132** facing inward. The side walls are formed by folding side panels **156** upward along the fold lines **112** as oriented in FIG. 8, and then folding the side panels **158** inward along fold lines **160** as indicated by the curved arrows and reference characters **1** in FIG. 8. Then the end flaps **142** can be folded inward as indicated in FIG. 9 by the rounded arrows and reference character **2**, as described above with reference to FIG. 4. Finally, the inward and outward facing surfaces of the end walls **116**, **122** are formed of the first treated surface **130** by folding the end panels **138** upward along the fold lines **118**, **124**, and folding the end panels **140** inward over the end flaps **142** along the double fold lines **162** to lock the tabs **144** in the slots **146** much as described above with respect to FIG. 5.

[0032] The methods and systems of the present disclosure, as described above and shown in the drawings, provide for manually forming containers from blanks, wherein a single surface of the blank forms interior and exterior facing surfaces of certain walls of the container. While the apparatus and methods of the subject disclosure have been shown and described with reference to preferred embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the scope of the subject disclosure.

Claims

1. A carton for storage and display of articles comprising: a bottom wall of a blank; a first side wall foldably connected to the bottom wall; a second side wall disposed opposite the first side wall and foldably connected to the bottom wall; a first end wall foldably connected to both the bottom wall and to the first and second side walls; and a second end wall disposed opposite the first end wall and foldably connected to both the bottom wall and to the first and second side walls, wherein each of the bottom wall, the first and second side walls, and the first and second end walls are configured to partially define an interior space; wherein one side of the blank defines a continuous surface partially defined by an interior-facing surface of the first side wall, an interior-facing surface of the second side wall, an exterior-facing surface of the first side wall, and an exterior-facing surface of the second side wall in a carton formed from the blank.
2. The carton as recited in claim 1, wherein the continuous surface is partially defined by an interior-facing surface of the bottom wall.
3. The carton as recited in claim 2, wherein the first end wall includes a first end panel connected along a fold line to the bottom wall, and a first end flap connected along a fold line to the first end panel.
4. The carton as recited in claim 3, wherein an exterior-facing portion of the first end panel is defined by a surface opposite the continuous surface, and wherein an interior-facing portion of the first end flap is defined by the surface opposite the continuous surface.
5. The carton as recited in claim 4, wherein end wall flaps of the first and second side wall are layered between the first end panel and the first end flap.
6. The carton as recited in claim 4, wherein one or more locking tabs extend from an end of the first end flap, wherein the one or more locking tabs is engaged in one or more respective locking slots in the bottom wall.
7. The carton as recited in claim 1, wherein the continuous surface is partially defined by an exterior-facing surface of the bottom wall.
8. The carton as recited in claim 7, wherein the first end wall includes a first end panel connected along a fold line to the bottom wall, and a first end flap connected along a fold line to the first end panel.
9. The carton as recited in claim 8, wherein an exterior-facing portion of the first end panel is defined by the continuous surface, and wherein an interior-facing portion of the first end flap is defined by the continuous surface.
10. The carton as recited in claim 9, wherein end wall flaps of the first and second side wall are layered between the first end panel and the first end flap.
11. The carton as recited in claim 9, wherein one or more locking tabs extend from an end of the first end flap, wherein the one or more locking tabs is engaged in one or more respective locking slots in the bottom wall.
12. The carton as recited in claim 1, further comprising: a first corner wall foldably connected to the first side wall, and to the first end wall; a second corner wall foldably connected to the first end wall, and to the second side wall; a third corner wall foldably connected to the second side wall, and to the second end wall; and a fourth corner wall foldably connected to the second end wall, and to the first side wall.
13. The carton as recited in claim 12, wherein the continuous surface further includes an interior-facing surface of the first corner wall, an interior-facing surface of the second corner wall, an interior-facing surface of the third corner wall, and an interior-facing surface of the fourth corner wall.
14. The carton as recited in claim 13, wherein the continuous surface further includes an exterior-facing surface of the first corner wall, an exterior-facing surface of the second corner wall, an

exterior-facing surface of the third corner wall, and an exterior-facing surface of the fourth corner wall.

15. The carton as recited claim 13, wherein the continuous surface is a treated surface and a second surface, opposite the continuous surface is an untreated surface, wherein the treated surface having received one or more surface treatments, and wherein the untreated surface not having received the one or more surface treatments.

16. A blank for forming a carton, the blank comprising: a first surface; a second surface disposed opposite the first surface; and a plurality of panel and flaps configured to form a carton defining a bottom wall, a first side wall foldably connected to the bottom wall, a second side wall disposed opposite the first side wall and foldably connected to the bottom wall, a first end wall foldably connected to the bottom wall, a second end wall disposed opposite the first end wall and foldably connected to the bottom wall, wherein each of the bottom wall, the first and second side walls, and the first and second end walls are configured to partially define an interior space, and wherein the first surface is configured to be at least partially defined by an interior-facing surface of the bottom wall, an interior-facing surface of the first side wall, an interior-facing surface of the second side wall, an exterior-facing surface of the first side wall, and an exterior-facing surface of the second side wall, in a carton formed from the blank.

17. The blank as recited in claim 16, wherein the plurality of flaps includes a first side wall panel foldably connected to a bottom panel, and a first side flap foldably connected to the first side wall panel, wherein each of the first side wall panel and the second side wall panel include respective laterally extending corner panels foldably connected thereto, wherein each of the respective laterally extending corner panels includes a respective end panel foldably connected thereto.

18. A method of forming a carton comprising: starting with a flat blank with a first treated surface on one side and a second surface on an opposite side from the first treated surface, manually forming a container from the flat blank, wherein manually forming the container includes: folding a first side wall panel upward along a fold line connecting the first side wall panel to a bottom panel; folding a second side wall panel downward along a fold line connecting the second side wall panel to the first side wall panel to form a first side wall with an interior-facing surface of the first side wall and an exterior facing surface of the first side wall both formed of the first treated surface; folding a lateral end flap of at least one of the first side wall panel and/or second side wall panel to form a portion of a first end wall; folding a first end panel along a fold line connecting the first end panel to the bottom panel and folding a first end flap along a fold line connecting the first end flap to the first end panel to enclose the lateral end flap between the first end panel and the first end flap; and securing the first end flap to the bottom panel to form the first end wall.

19. A method as recited in claim 18, wherein manually forming the container from the blank includes forming the bottom panel into a bottom wall of the container wherein the first treated surface is on an inward facing surface of the bottom wall, and wherein inward and outward facing surfaces of the first end wall are formed of the second surface of the blank.

20. A method as recited in claim 18, wherein manually forming the container from the blank includes forming the bottom panel into a bottom wall of the container wherein the second surface is on an inward facing surface of the bottom wall, and wherein inward and outward facing surfaces of the first end wall are formed of the first treated surface.
